

3. Prototype / Implementation Plan

To transition the current GitHub MVP into a production-grade system, we will execute a 3-Phase Rollout Strategy:

Phase 1: MVP Hardening & Live Pilot (Months 1-2)

- **Objective:** Replace mock data with live API integrations and validate the model in a real-world scenario.
- **Action Items:**
 - Deprecate `mock_data_generator.py` and implement production-ready API connectors for IMD, NDMA, and Bhuvan GIS.
 - Deploy the existing Dockerized stack to a managed cloud environment (e.g., AWS EC2 or Google Cloud Run).
 - Run a localized pilot with 1-2 partner NGOs in coastal Odisha during the pre-monsoon season to validate the `predict.py` accuracy.

Phase 2: Platform Scaling & Security (Months 3-5)

- **Objective:** Prepare the platform for multi-tenant NGO usage.
- **Action Items:**
 - Migrate the local PostgreSQL database to a highly available Managed Cloud SQL instance.
 - Implement Role-Based Access Control (RBAC) in Streamlit so different NGOs can securely connect their private warehouse inventory data without cross-exposure.
 - Integrate SMS/WhatsApp API (Twilio/Gupshup) to send automated route manifests directly to truck drivers.

Phase 3: Nationwide Expansion & Advanced Capabilities (Months 6-12)

- **Objective:** Scale beyond Odisha and introduce edge capabilities.
- **Action Items:**
 - Expand geographic coverage and model training to include flood models for Assam, Bihar, and Kerala.
 - Integrate ISRO satellite imagery with Computer Vision pipelines for real-time flood inundation mapping.

- Launch a lightweight React Native mobile app for on-ground volunteers to provide real-time feedback, creating a closed-loop system to retrain the ML models.